COMPENDIUM OF INDIAN STANDARDS

RENEWABLE ENERGY (WIND TURBINES AND MARINE ENERGY CONVERSION SYSTEMS)

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Introduction

Renewable Energy sector is pivotal in driving the global transition towards a sustainable future. By replacing fossil fuels with clean sources like solar, wind, marine and hydropower, renewable energy significantly reduces greenhouse gas emissions, helping to mitigate climate change and promote environmental sustainability. At COP26 in Glasgow, India demonstrated its strong commitment through the **"Panchamrit"** strategy, which includes achieving 500 GW of non-fossil energy capacity, meeting 50% of energy needs from renewables, reducing carbon emissions by one billion tonnes, cutting carbon intensity by 45% by 2030, and attaining net-zero emissions by 2070. These goals highlight India’s leadership in climate action and sustainable development. Wind turbine power plants play a critical role in India’s renewable energy roadmap. They offer a clean, sustainable, and cost-effective solution to meet the rising power demands of a growing population while mitigating the environmental impacts associated with conventional energy generation. To promote standardization in the **wind energy sector**, standards have been developed covering various aspects such as wind turbine design, power performance measurement, mechanical load assessment, acoustic noise, and power quality. These standards ensure that wind turbines operate efficiently, safely, and reliably under diverse environmental conditions. Safety remains a key focus area, with specific standards addressing structural integrity, electrical safety, and protection against hazards such as mechanical failures and lightning strikes. By harmonizing design, testing, and performance criteria, these standards support the safe deployment, grid integration, and long-term sustainability of wind energy systems.

India, endowed with a vast coastline and diverse topography, possesses significant wind energy potential. Standardization in the **marine energy** domain focuses on critical areas such as terminology, resource assessment, design and safety, system performance, environmental survivability, and grid integration. These standards provide a framework to guide the development, testing, deployment, operation, and decommissioning of marine energy systems.

1. Standards for Components of Wind Turbine

a. IS 16589 (Part 4) : 2017 Design requirements for wind turbine gearboxes

This standard applies to enclosed speed-increasing gearboxes used in horizontal axis wind turbines with power ratings above 500 kW, whether installed onshore or offshore. It provides guidance on analyzing wind turbine loads for gearbox and gear element design, covering spur, helical, and double helical gears in both parallel and epicyclic configurations. The standard primarily addresses gearboxes with rolling element bearings, though plain bearings may be used, their design and rating are not included. It also offers recommendations for the integrated design of shafts, bearings, shaft-hub interfaces, and gear case structures to ensure durability under operational conditions. Additionally, it includes provisions for lubrication, prototype and production testing, and the gearbox's operation and maintenance.

b. IS/IEC 61400-2 : 2013 Small Wind Turbines (SWT)

This part of IS/IEC 61400 specifies safety, quality assurance, and engineering integrity requirements for small wind turbines (SWTs) with rotor swept areas ≤200 m² and voltage below 1000 V AC or 1500 V DC. It covers design, installation, maintenance, and operation, ensuring protection from hazards throughout the turbine's lifetime.

2. Code of Practice for Wind Turbine Lightning Protection (IS/IEC 61400-24)

This standard addresses lightning protection for wind turbine generators and wind power systems. It outlines the lightning environment, risk assessment, and protection requirements for blades, structural components, and electrical/control systems. It includes compliance test methods, references to relevant standards, personal safety guidance, and damage reporting procedures.

3. Standards for Power Performance Measurement of Wind Turbines

a. IS/IEC 61400- 12-1(Electricity Producing Wind Turbines - Power performance measurements)

These standard outlines procedures to measure power performance characteristics of wind turbines of all sizes connected to power grids or battery banks. It includes determining power curves and annual energy production (AEP), considering wind speed and turbine output, and emphasizes site-specific data, performance comparisons, and assessment of measurement uncertainties.

b. IS/IEC 61400-12-3(Power performance - Measurement based site calibration)

This standard provides a procedure to measure and analyze wind speed corrections due to terrain effects, essential for accurate performance testing of wind turbines of all types and sizes. The procedure applies to the performance evaluation of specific wind turbines at specific locations.

c. IS/IEC 61400-12-4(Power performance - Numerical site calibration for power performance testing of wind turbines)

This standard outlines the current state of numerical flow modelling over terrain for wind applications. It highlights key technical aspects, existing guidelines, and open issues, while recommending further validation through benchmarking to improve accuracy and reliability in wind energy assessments.

d. IS/IEC 61400-12-5(Power performance - Assessment of obstacles and terrain)

These standard outlines procedures to identify measurement sectors unaffected by turbines or obstacles for accurate wind assessments. It supports power performance evaluations, nacelle power curve and transfer function determination, and terrain classification. These procedures help estimate uncertainty and decide if site calibration is necessary based on terrain complexity.

e. IS/IEC 61400-12-6(Measurement based nacelle transfer function of electricity producing wind turbines)

These standard details a procedure to measure the nacelle transfer function of large horizontal-axis wind turbines. It addresses rotor-induced wind speed distortion at the nacelle, applies correction methods, and emphasizes uncertainty assessment. Simultaneous nacelle and free-stream wind speed measurements ensure reliable characterization over varied wind conditions.

4. Standards for Communications, Monitoring and Control of Wind Turbines

a. IS/IEC 61400- 25-1 - Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 1 Overall description of principles and models

The standard provides an introduction to communication in wind power plants using a client-server model. It defines three key components: wind power plant information models, information exchange models, and their mapping to communication profiles. The standard ensures interoperability between diverse systems by offering a uniform, scalable, component-based communication framework.

b. IS/IEC 61400- 25-2 - Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 2 Information model

This standard defines the information model for wind power plant devices and functions, specifying logical node names and data names for component communication. It details relationships between logical devices and nodes, and common data classes such as setpoints, status, alarms, commands, events, timing, and alarm statuses.

c. IS/IEC 61400- 25-3 - Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 3 Information exchange models

This standard defines the information model for wind power plant devices and functions, specifying logical node names and data names for component communication. It details relationships between logical devices and nodes, and common data classes such as setpoints, status, alarms, commands, events, timing, and alarm statuses.

d. IS/IEC 61400- 25-4- Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 4 Mapping to communication profile

This standard defines test cases for conformance testing of servers and clients, specifying protocol stack mappings for communication. It covers message encoding for data access, device control, event reporting, publisher/subscriber functions, device self-description, and data typing, ensuring reliable client-server information exchange.

e. IS/IEC 61400- 25-5 - Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 5 Compliance testing

This standard specifies standard methods and test cases for compliance testing of client and server devices in wind power plants. It defines measurement metrics to ensure devices integrate smoothly, operate correctly, and meet communication requirements, enhancing system reliability and performance for users.

f. IS/IEC 61400- 25-6 - Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 6 Logical node classes and data classes for condition monitoring

This standard defines an information model for wind turbine condition monitoring, extending IS 61400-25-2 to describe and exchange sensor and calculated data. It introduces a multidimensional operational state bin to compare complex conditions, supporting predictive maintenance through parameters like vibration, oil debris, temperature, strain, and acoustic analysis.

g. IS/IEC 61400- 25-71 - Wind energy generation systems - Part 25-71: Communications for monitoring and control of wind power plants - Configuration description language

This standard defines an information model for wind turbine condition monitoring using IS 61400-25-2 extensions. It introduces a multidimensional operational state bin to classify complex conditions, supporting predictive and condition-based maintenance. It covers sensor and calculation data like vibration, oil debris, temperature, strain, and acoustic analyses for effective monitoring.

h. IS/IEC 61400- 26-1- Wind Energy Part 26 Generation Systems Section 1 Availability

This standard establishes a common information model for exchanging availability metrics among wind power stakeholders, supporting energy production and asset management. It defines time-based and production-based availability indicators, standardizes calculation methods, and provides clear data categories with priority rules to ensure consistent, unambiguous reporting across users.

5. Standards for - Electrical simulation models and Testing of Wind Turbines

a. IS/IEC 61400- 27-1 - Wind energy generation systems - Part 27: Electrical simulation models - Section 1: Generic models

This standard defines time-domain electrical simulation models for wind turbines and power plants, used in power system stability analyses. It specifies generic parameters and modular models for current and future turbine and plant configurations, enabling dynamic simulation of electrical characteristics for stability studies in power systems.

b. IS/IEC 61400- 27-2- Wind energy generation systems - Part 27: Electrical simulation models - Section 2: Model validation

This standard outlines procedures for validating electrical simulation models of wind turbines and power plants for use in grid stability analyses. Based on IS 61400-21 tests, it covers fault ride-through, control performance (power, frequency, inertia, reactive), and applies to both turbine terminals and power plant points of connection.

c. IS/IEC 61400- 21-1 - Wind Energy Generation Systems Part 21 Measurement and Assessment of Electrical Characteristics Section 1 Wind Turbines

This standard defines quantities, measurement procedures, and assessment methods for evaluating the electrical characteristics of grid-connected wind turbines. It focuses on single turbines with three-phase connections, ensuring power quality and compliance with grid requirements, using standardized, site-independent testing methods applicable to stable grid frequency conditions.

d. IS/IEC 61400- 21-3 - Wind energy generation systems - Part 21: Measurement and assessment of electrical characteristics - Section 3: Wind turbine harmonic model and its application

This standard provides a standardized methodology for developing and validating harmonic models of grid-connected wind turbines. It ensures consistency and accuracy in harmonic analysis, supporting stakeholders in design, evaluation, and mitigation of harmonics in onshore and offshore wind power plants, particularly in complex, multi-stakeholder grid integration scenarios.

e. IS 16589 :Part 11 (Acoustic Noise Measurement Techniques)

This standard outlines standardized procedures for accurately measuring noise emissions from individual wind turbines near the source, minimizing propagation errors. Unlike community noise assessments, it focuses on noise characterization across varying wind speeds and directions, enabling consistent evaluation and comparison between different wind turbine models.

g. IS/IEC 61400- 23- Wind Turbines Part 23 Full-Scale Structural Testing of Rotor Blades

This standard specifies requirements for full-scale structural testing of wind turbine blades, including static and fatigue tests, to assess blade integrity. It ensures blades meet design assumptions with high confidence. The standard supports manufacturers and third-party evaluators in verifying blade performance under realistic loading conditions using design loads and material data.

h. IS/IEC/TS 61400-13 - Wind Turbines Part 13: Measurement of Mechanical Loads

The standard outlines procedures for measuring mechanical loads on wind turbines to validate load simulation models. It covers site selection, data acquisition, calibration, and reporting. Though primarily for large onshore horizontal-axis turbines, the methods may apply to other turbine types and serve additional testing and safety purposes.

i. IS/IEC 61400- 50- Wind energy generation systems - Part 50: Wind measurement - Overview

This standard provides standardized methodologies for accurate and consistent wind measurement, supporting various applications like power performance and noise assessment. It covers measurement methods, equipment, calibration, and uncertainty analysis.

j. IS/IEC 61400- 50-3 - Wind energy generation systems - Part 50-3: Use of nacelle-mounted lidars for wind measurements

This standard outlines standardized procedures for consistent and high-quality wind measurements using forward-looking nacelle-mounted wind lidars. Applicable to all lidar types, it ensures best practices regardless of instrument model or measurement principle, supporting accurate data collection specifically for power performance testing of wind turbines.

List of Standards related to Wind Energy Sector

Sl. No.	IS No.	Title
1.	IS 16589 (Part 4) : 2017 IEC 61400-4 : 2012	Wind turbines: Part 4 design requirements for wind turbine gearboxes
2.	IS 16589 (Part 11) : 2018 IEC61400-11 : 2012	Wind Turbines Part 11 Acoustic Noise Measurement Techniques
3.	IS/IEC 61400-2 : 2013 IEC 61400-2:2013	Wind Turbines Part 2: Small Wind Turbines
4.	IS/IEC 61400-12-1 : 2017	Wind Energy Generation Systems Part 12 Electricity Producing Wind Turbines Section 1 Power performance measurements
5.	IS/IEC 61400-12-3) : 2022 IEC 61400-12-3:2022	Wind energy generation systems - Part 12: Power performance - Section 3: Measurement based site calibration
6.	IS/IEC 61400-12-4) : 2020 IEC TR 61400-12-4:2020	Wind energy generation systems - Part 12- Power performance - Section 4 Numerical site calibration for power performance testing of wind turbines
7.	IS/IEC 61400-12-5) : 2022 IEC 61400-12-5:2022	Wind energy generation systems - Part 12: Power performance - Section 5: Assessment of obstacles and terrain
8.	IS/IEC 61400-12-6) : 2022 IEC 61400-12-6:2022	Wind energy generation systems - Part 12-6: Measurement based nacelle transfer function of electricity producing wind turbines
9.	IS/IEC/TS 61400-13 : 2015 IEC 61400 : Part 13	WIND TURBINES PART 13: MEASUREMENT OF MECHANICAL LOADS
10.	IS/IEC/TS 61400-14 : 2005 IEC 61400-14:2005	Wind Turbines Part 14: Declaration of Apparent Sound Power Level and Tonality values
11.	IS/IEC 61400-21-1) : 2019 IEC 61400-21-1:2019	Wind Energy Generation Systems Part 21 Measurement and Assessment of Electrical Characteristics Section 1 Wind Turbines
12.	IS/IEC/TR 61400-21-3) : 2019 IEC TR 61400-21-3:2019	Wind energy generation systems - Part 21: Measurement and assessment of electrical characteristics - Section 3: Wind turbine harmonic model and its application
13.	IS/IEC 61400-22 : 2010 IEC 61400-22 : 2010	Wind Turbines Part 22 Conformity Testing and Certification
14.	IS/IEC 61400-23 : 2014 IEC 61400-23 : 2014	Wind Turbines Part 23 Full-Scale Structural Testing of Rotor Blades
15.	IS/IEC 61400-24 : 2019 IEC 61400-24:2019	Wind energy generation systems - Part 24: Lightning protection
16.	IS/IEC 61400-25-1) : 2017 IEC 61400-25-1 : 2017	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 1 Overall description of principles and models
17.	IS/IEC 61400-25-2) : 2015 IEC 61400-25-2 : 2015	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 2 Information model
18.	IS/IEC 61400-25-3) : 2015 IEC 61400-25-3 : 2015	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 3 Information exchange models

19.	IS/IEC 61400-25-4) : 2016	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 4 Mapping to communication profile
20.	IS/IEC 61400-25-5) : 2017 IEC61400-25-5 : 2017	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 5 Compliance testing
21.	IS/IEC 61400-25-6) : 2016	Wind Turbines Part 25 Communications for Monitoring and Control of Wind Power Plants Section 6 Logical node classes and data classes for condition monitoring
22.	IS 61400 (Part 25/Sec 71) : 2019 IECTS 61400-25-71:	Wind energy generation systems - Part 25-71: Communications for monitoring and control of wind power plants - Configuration description language
23.	IS/IEC 61400-26-1) : 2019 IEC 61400-26-1:2019	Wind Energy Part 26 Generation Systems Section 1 Availability
24.	IS/IEC 61400-27-1) : 2020 IEC 61400-27-1:2020	Wind energy generation systems - Part 27: Electrical simulation models - Section 1: Generic models
25.	IS/IEC 61400-27-2) : 2020 IEC 61400-27-2:2020	Wind energy generation systems - Part 27: Electrical simulation models - Section 2: Model validation
26.	IS/IEC 61400-50 : 2022 IEC 61400-50:2022	Wind energy generation systems - Part 50: Wind measurement - Overview
27.	IS/IEC 61400-50-3) : 2022 IEC 61400-50-3:2022	Wind energy generation systems - Part 50-3: Use of nacelle-mounted lidars for wind measurements

Standards related to Marine Energy Conversions Systems

1. Marine Energy — General Terms, Design Requirements and General Guidance

a. IS 18972 (Part 1): 2025 - Marine Energy – Wave, Tidal, and Other Water Current Converters – Part 1: Vocabulary

This standard defines standardized terminology for marine energy, focusing on devices that convert wave, tidal, and water current energy into electricity. It excludes terms related to dams, tidal barrages, offshore wind, and other sources, aiming to ensure consistent communication within the marine energy sector.

b. IS 18972 (Part 2): 2025 - Marine Energy – Wave, Tidal, and Other Water Current Converters – Part 2: Marine Energy Systems – Design Requirements.

This standard outline design requirements for marine energy converters (MECs) to ensure structural and operational integrity under site-specific conditions. It addresses hazards, limit states, and design categories for floating or fixed devices using wave, tidal, or river currents, supporting safe development, deployment, and certification of MEC systems.

c. IS 18972 (Part 20): 2025 - Marine Energy – Wave, Tidal, and Other Water Current Converters – Part 20: Design and Analysis of an Ocean Thermal Energy Conversion (OTEC) Plant – General Guidance.

This standard provides general design and assessment principles for Ocean Thermal Energy Conversion (OTEC) plants, covering land-based, shelf-mounted, and floating systems. It focuses on OTEC-specific components, especially warm and cold water intake, guiding developers, engineers, and financiers for stable power generation and grid connection.

2. Marine Energy –Assessment of Mooring System for Marine Energy Converters (MECS)

a. IS 18972 (Part 10): 2025 - Marine Energy – Wave, Tidal, and Other Water Current Converters – Part 10: Assessment of Mooring System for Marine Energy Converters (MECS).

The Standard provide uniform methodologies for the design and assessment of mooring systems for floating Marine Energy Converters (MECs) . It is intended to be applied at various stages, from mooring system assessment to design, installation and maintenance of floating Marine Energy Converters plants.

3. Marine Energy – Electricity Producing Wave Energy converters – Power Performance Assessment

a. IS 18972 (Part 100): 2025 - Marine Energy – Wave, Tidal, and Other Water Current converters – Part 100: Electricity Producing Wave Energy Converters – Power Performance Assessment.

This standard, provides a method for assessing the electrical power production performance of a Wave Energy Converter (WEC), based on the performance at a testing site.